Anderson's Bakery Management System (LogiSyn)

Comprehensive User Guide & Technical Manual

Empowering Factory Floor Operations with Simple, Automated, and Resilient Order Management

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1. Introduction & Overview

LogiSyn is a production management system engineered specifically for Anderson's Bakery in Cape Town. Anderson's supplies fresh, chilled, ambient, and frozen baked goods to leading retail supermarkets including Checkers and Spar.

The Core Problem LogiSyn Solves

Historically, incoming customer purchase orders were transferred by hand into multiple disconnected physical logbooks and spreadsheets. This led to:

- Calculation bottlenecks when converting retail loaf and pastry counts into batch ingredients.
- Human error in pan and trolley allocations, resulting in production delays or wasted oven space.
- Discrepancies between planned raw materials and actual bakery floor usage.
- High vulnerability to internet drops and load-shedding disruptions.

How LogiSyn Works

LogiSyn transforms this workflow into an automated four-step process:

1. **Ingest Order:** Drag-and-drop an incoming customer order (PDF or Excel).
2. **Auto-Scale:** LogiSyn automatically reads the customer, line items, and quantities, calculating exact pans, trolleys, and recipe ingredient totals (including a 10% safety buffer).
3. **Bake & Track:** Bakers on the floor view the digital production sheet, record actuals, or upload a filled production Excel file.
4. **Complete & Archive:** Submitting the order marks it complete, persists the records across local storage, SQL Server LocalDB, and cloud MongoDB Atlas, and moves it to searchable historical archives.

2. User Roles & Permissions

LogiSyn implements strict Role-Based Access Control (RBAC). The interface adapts automatically to the user's role upon login, ensuring floor staff are not overwhelmed with administrative settings while administrators retain full governance over products and user access.

Screen / Feature	Admin	Manager	User (Baker / Floor)
Dashboard	**Full access (KPIs + Export/Email)**	View KPIs & Recent Orders	View KPIs & Personal Shift
Orders View	Full table, search, filters & create	Full table, search, filters & create	Streamlined active production list
Order Ingestion (PDF/Excel)	**Yes**	**Yes**	No
Production Sheets Breakdown	Read & Summary Paper	Detailed & Editable Sheet	Replaced by User Order Sheet
User Production Sheet	Read-only	Read & Edit	Interactive (Actuals & complete)
Excel Production Upload	Managed via Ingestion	Managed via Ingestion	Upload completed floor sheet
Order History	Full search & date filters	Full search & date filters	Hidden
Product & Recipe Catalog	**Full CRUD (Add/Edit/Delete)**	**Full CRUD (Add/Edit/Delete)**	Hidden
User Administration	Full CRUD & Cloud Sync	Hidden	Hidden
Excel & Email Reporting	Full access from Dashboard	Hidden on Dashboard	Hidden

3. System Access & Getting Started

Launching the Desktop Application

1. Double-click the LogiSyn shortcut on your Windows desktop, or launch LogiSyn.exe from your installation folder.
2. The Login Screen will appear: enter your assigned Username and Password, optionally click the Eye Icon to toggle masking, and press Login.

Default Seed Accounts

Role	Username	Initial Password	Recommended Purpose
Administrator	admin	admin123	System setup, user administration, reporting, recipe configuration
Manager	manager	manager123	Order scheduling, recipe review, daily production oversight
User (Baker)	user	user123	Factory floor baking, pan allocations, order completion

> **[SECURITY NOTICE]** Automatic Password Security Upgrade: When an account logs in for the first time, LogiSyn

> automatically upgrades plain-text passwords to secure, salted PBKDF2-HMAC-SHA256 hashes with 600,000 iterations. Change default passwords after initial deployment.

Understanding the Connection Status Badge

In the top-right corner of the application header, an intelligent connection badge indicates live connectivity:

- **API Connected (Green):** Online with Render Cloud API & MongoDB Atlas. Orders and user updates sync immediately to cloud and local storage.
- **Calling API... (Blue):** An in-flight request is communicating with cloud services. The application remains fully responsive via background threads.
- **Local Storage / Fallback (Orange):** Offline mode (load-shedding, internet down, or server sleeping). Zero downtime; app operates seamlessly against local SQL Server LocalDB and JSON files.

4. Step-by-Step Module Guides

Module 1: Dashboard Overview

The central command center provides real-time counts of New/Pending and Completed Orders, a chronological recent orders feed, and quick-action shortcuts for Admin users (Export to Excel and Email Orders).

Module 2: Order Ingestion & Automatic Scaling

Ingest customer orders via PDF, XLSX, XLS, or CSV. When processed, LogiSyn extracts customer data, order number, and line items, then runs the auto-scaling engine:

• Pans Required: ceil(Quantity / 50)
• Trolleys Required: ceil(Quantity / 100)
• Raw Materials: Base Recipe Qty × Order Qty × 1.10 (incorporating a 10% safety buffer)
• Collision Detection: Automatically appends incremental suffixes (e.g., #CK-90214-1) if an ID already exists.

Module 3: Production Sheets & Manager Order Review

Presents a dual-panel review: the left panel outlines line items, pan/trolley allocations, and baking notes; the right panel consolidates bakery-wide raw materials. Managers can add products on the fly, export to Excel, or commit to the active floor queue.

Module 4: Factory Floor Production Workflow (Baker)

Bakers log in to view active pending orders. They record actual raw materials used, pans, trolleys, and operational notes. Clicking 'Mark as Complete' locks the sheet to read-only and archives it. Alternatively, staff can print or export to Excel, track actuals on paper/spreadsheet, and re-upload the completed file via 'Upload Production Excel Sheet'.

Module 5: Order History & Quality Audit

Admins and Managers can search past batches by customer or PO number, filter by date ranges (Today, This Week, This Month, All), and inspect read-only production logs for compliance audits.

Module 6: Product & Recipe Catalog Management

Configure bakery items, prices, storage categories (Ambient, Chilled, Frozen), shelf-life limits (Sell By and Best Before days), mixing/baking instructions, and itemized recipe ingredients with precise units.

Module 7: User Management & Cloud Sync

Manage team credentials and assign roles (Admin, Manager, User). When users are created during offline periods or load-shedding, they are tagged as local orphans. Clicking the 'Sync' button pushes all pending accounts to MongoDB Atlas once internet is restored.

Module 8: Reporting, Excel Export & Email Dispatch

Generate multi-sheet OpenXML workbooks summarizing batches and raw material usage. Administrators can dispatch reports directly through Outlook COM automation or standard mailto links.

5. Offline Resilience & Load-Shedding Survival

LogiSyn features a Local-First, Failover Hybrid Strategy tailored for uninterrupted production in South Africa:

- Local-First Order Creation: Saves locally to SQL Server LocalDB and JSON stores first before cloud sync.
- Zero-Cloud Parsing: Uses native libraries (PdfPig and ClosedXML) running entirely in-process on the local PC.
- Silent Offline Fallback: Seamlessly redirects queries to local storage if Render or MongoDB Atlas cannot be reached.
- Automatic Reconciliation: Background pingers and manual Sync buttons push local changes to the cloud when connectivity returns.

6. Quick Reference Cheat Sheet

Task	Steps	Location
Log In	Enter Username & Password -> click Login	Login Screen
Ingest New Order	Orders -> Create Order -> Drag/Drop PDF -> Go	Orders View (Top Right)
Add Extra Product	Inside Order Sheet -> click Add Product	Order Sheets View
Save Production Excel	Inside Order Sheet -> click Save Excel	Order Sheets View
Finish Order (Baker)	User Orders -> open Order -> enter actuals -> Mark Complete	User Order Sheet View
Upload Floor Excel	User Orders -> click Upload Production	User Orders View

	Excel Sheet	
Search Past Orders	History -> type in search bar or select Date filter	History View
Add New Recipe	Products -> Add Product -> enter method & ingredients -> Save	Manage Products View
Sync Offline Users	Users -> click Sync button	Manage Users View
Test Cloud Status	Click the Connection Pill in header	Application Header

